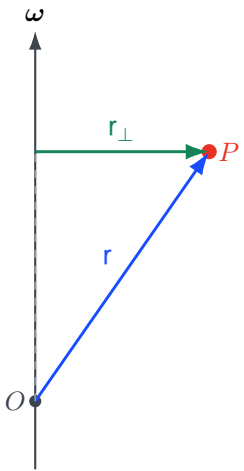
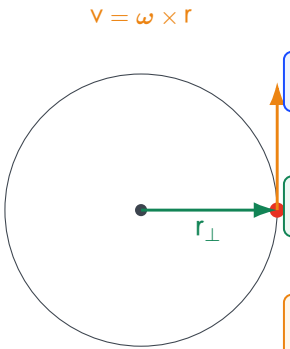
定轴转动质点的角动量：一般不平行于转轴，特殊横截面内才同向



含轴向分量时，L 一般有横向分量



垂直转轴观察的横截面

$$\mathbf{L} = \mathbf{r} \times m(\boldsymbol{\omega} \times \mathbf{r})$$

$$\mathbf{L} = m[r^2\boldsymbol{\omega} - (\mathbf{r} \cdot \boldsymbol{\omega})\mathbf{r}]$$

$\mathbf{r} \perp \boldsymbol{\omega} \Rightarrow \mathbf{L} = mr^2\boldsymbol{\omega}$；
$r = 0$ 或 $\omega = 0$ 时 $\mathbf{L} = 0$，无“同向”可言。